

Bloc D — Fiche express

L'intégrale de Riemann: **7 families**, their **signals**, their **steps**

Worth 25–40 pts / 100. Families 1–4 alone = 20–25 pts every session.

0. Decision in 5 seconds — look at what is IN FRONT of the integral

Read the limit, match one row, apply the tool

You see	Family	Tool	Answer looks like
$g(x) = \int_{v(x)}^{u(x)} f$, “calculer g' et domaine”	F1	HAUT – BAS	$u'f(u) - v'f(v)$ + an interval
$\lim_{x \rightarrow x_0} \frac{1}{x-x_0} \int_a^{h(x)} f$	F2	nombre dérivé	$g'(x_0)$, one number
$\lim_{n \rightarrow \infty} \sum_{k=1}^n (\dots k/n \dots)$	F3	somme de Riemann	$\int_0^1 f$, often with $\text{Ln } 2$
$\lim_{n \rightarrow \infty} \left(\prod_{k=1}^n \dots \right)^{1/n}$ or $\sqrt[n]{\prod}$	F4	Ln , then Riemann	$e^{\text{something}}$
$\lim_{x \rightarrow 0^+} \int_x^{2x} \frac{h(t)}{t} dt$	F5a	th. de la moyenne	$h(0) \cdot \text{Ln } 2$
$\lim_{x \rightarrow +\infty} \int_x^{2x} (\text{convergent})$	F5b	critère de Cauchy	0
$\int_0^{+\infty} \frac{f(ax)-f(bx)}{x} dx$ (guided, 15 pts)	F6	Frullani	$\ell \text{Ln } \frac{a}{b}$
“Montrer que f intégrable ssi $U - L < \varepsilon$ ”	F7	proof by heart	6 lines

The 4 mistakes that cost the most points

1. F4: forgetting the final $e^{\dots}$ (you computed $\text{Ln } L$, not L). **2.** F1: forgetting u' and v' in HAUT – BAS. **3.** F1 domain: testing the two bounds separately instead of the **whole segment**. **4.** F2: using L'Hôpital where “nombre dérivé” gives the answer in one line.

1. Formulas to know by heart (nothing else is needed)

HAUT – BAS (F1, F2)

Nombre dérivé (F2)

$$G(x) = \int_{v(x)}^{u(x)} f(t) dt \Rightarrow G'(x) = u'(x) f(u(x)) - v'(x) f(v(x))$$

$$\lim_{x \rightarrow x_0} \frac{g(x) - g(x_0)}{x - x_0} = g'(x_0)$$

Left: if the low bound is a constant, $v' = 0$ and only HAUT stays. Right: works only if $g(x_0) = 0$ (the two bounds coincide at x_0).

Somme de Riemann (F3, F4)

Théorème de la moyenne (F5a)

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f\left(\frac{k}{n}\right) = \int_0^1 f(x) dx$$

$$\int_x^{2x} \frac{h(t)}{t} dt = h(\xi_x) \text{Ln } 2, \quad \xi_x \in [x, 2x]$$

Left: pull out $\frac{1}{n}$, write the rest with $\frac{k}{n}$ only. Right: because $\int_x^{2x} \frac{dt}{t} = \text{Ln}(2x) - \text{Ln } x = \text{Ln } 2$; bounds $x, 3x$ give $\text{Ln } 3$.

The 4 reference integrals (they are the F4 answers)

$$\begin{aligned} \int_0^1 \text{Ln}(1+x) dx &= 2 \text{Ln } 2 - 1 & \int_0^1 \text{Ln}(1+x^2) dx &= \text{Ln } 2 - 2 + \frac{\pi}{2} \\ \int_0^1 \sqrt{1-x} dx &= \frac{2}{3} & \int_0^1 \sqrt{1-x^2} dx &= \frac{\pi}{4} \text{ (area of a quarter circle)} \end{aligned}$$

Primitives and values that appear every time

$$\int \operatorname{Ln} s \, ds = s \operatorname{Ln} s - s \qquad \int \frac{dt}{t} = \operatorname{Ln} |t| \qquad \int \frac{dx}{1+x^2} = \arctan x$$

$$\int \frac{u'}{u} = \operatorname{Ln} |u| \qquad \int u' u^n = \frac{u^{n+1}}{n+1} \qquad \int \frac{u'}{1+u^2} = \arctan u$$

$$\arctan 1 = \frac{\pi}{4}, \quad \operatorname{Ln} 1 = 0, \quad e^0 = 1, \quad \cos \pi = -1, \quad \sin \pi = 0, \quad \arcsin x + \arccos x = \frac{\pi}{2}.$$

Ln rules (F4 and Frullani)

$$\operatorname{Ln}(ab) = \operatorname{Ln} a + \operatorname{Ln} b \qquad \operatorname{Ln} \frac{a}{b} = \operatorname{Ln} a - \operatorname{Ln} b \qquad \operatorname{Ln}(a^p) = p \operatorname{Ln} a \qquad \operatorname{Ln} e^a = a$$

Consequences: $\operatorname{Ln} \prod a_k = \sum \operatorname{Ln} a_k$ (product becomes sum) and $\operatorname{Ln}(\dots)^{1/n} = \frac{1}{n} \operatorname{Ln}(\dots)$ (the $1/n$ exponent becomes the $\frac{1}{n}$ factor).

2. Family 1 — $g'(x)$ and its domain

6 questions, 3–5 pts each

SIGNAL: the question says “calculer $g'(x)$ et préciser le domaine” with $g(x) = \int_{v(x)}^{u(x)} f(t) \, dt$

Steps

1. Identify $f(t)$, $u(x)$ (HAUT), $v(x)$ (BAS).
2. Compute u' , v' .
3. Write $g'(x) = u' f(u) - v' f(v)$. [2 pts]
4. Domain D where f is **continuous**.
5. Require the **whole segment** between $v(x)$ and $u(x)$ to be inside D . [2–3 pts]
6. If D is two separate pieces, both bounds must be in the **same piece**.

Example 1.2 (Août 2020) — the domain trap

$$g(x) = \int_{x^2-1}^{x^2} \operatorname{Ln} t \, dt$$

$$g'(x) = 2x \operatorname{Ln}(x^2) - 2x \operatorname{Ln}(x^2 - 1)$$

$\operatorname{Ln} t$ needs $t > 0$. The whole segment $[x^2 - 1, x^2]$ must sit inside $]0, \infty[$, so its **lower** end $x^2 - 1 > 0$, i.e. $|x| > 1$.
Domain: $] - \infty, -1[\cup]1, +\infty[$.

Example 1.3 (Juil. 2021) — two pieces

$$g(x) = \int_x^{x-1} \operatorname{Ln}(\operatorname{Ln} |t|) \, dt, \quad f \text{ needs } |t| > 1.$$

$D =] - \infty, -1[\cup]1, \infty[$ is two pieces. The segment from $x - 1$ to x must sit in ONE piece:

right piece: $x - 1 > 1$, so $x > 2$; left piece: $x < -1$.

Domain: $] - \infty, -1[\cup]2, +\infty[$.

Wrong answer $]1, 2]$ comes from testing the bounds separately.

Example 1.1 (Oct. 2022)

$$g(x) = \int_0^{\sin x} e^{t^2} \, dt.$$

$$u = \sin x, \quad u' = \cos x, \quad v = 0.$$

$$g'(x) = \cos x \cdot e^{\sin^2 x}.$$

$$f \text{ continuous on } \mathbb{R} \Rightarrow \text{domain} = \mathbb{R}.$$

3. Family 2 — limit that is a nombre dérivé

3 questions, 3 pts each

SIGNAL: $\lim_{x \rightarrow x_0} \frac{1}{x - x_0} \int_a^{h(x)} f(t) \, dt$ (a division by $x - x_0$ in front, one bound fixed)

Steps (5 lines on the exam)

1. **Name it:** $g(x) = \int_a^{h(x)} f(t) \, dt$.
2. **Check zero:** $h(x_0) = a$ so $g(x_0) = \int_a^a = 0$. [1 pt]
3. **Recognize:** $\frac{g(x)-0}{x-x_0} \rightarrow g'(x_0)$.
4. **Compute:** $g'(x) = h'(x) f(h(x))$ (HAUT – BAS). [2 pts]
5. **Plug in** $x = x_0$.

Example 2.1 (Oct. 2022)

$$\lim_{x \rightarrow \pi} \frac{1}{x - \pi} \int_0^{\sin x} e^{t^2} \, dt$$

$$g(x) = \int_0^{\sin x} e^{t^2} \, dt; \quad \sin \pi = 0 \text{ so } g(\pi) = 0.$$

$$g'(x) = \cos x e^{\sin^2 x}, \text{ so the limit is } g'(\pi) = \cos \pi \cdot e^0 = \boxed{-1}.$$

Example 2.2 (Juil. 2023)

$$\lim_{x \rightarrow 1} \frac{1}{x-1} \int_0^{\text{Ln } x} f(t) dt, \quad f(t) = \frac{\sin t}{t}, \quad f(0) = 1$$

$\text{Ln } 1 = 0$ so $g(1) = 0$. $g'(x) = \frac{1}{x} f(\text{Ln } x)$, so the limit is $g'(1) = 1 \cdot f(0) = \boxed{1}$.

Example 2.3 (cours)

$$\lim_{x \rightarrow 0} \frac{1}{x} \int_0^x f(t) dt, \quad f \text{ continuous}$$

$$F(x) = \int_0^x f, \quad F(0) = 0, \quad F' = f. \quad \text{The limit is } F'(0) = \boxed{f(0)}.$$

Why NOT the moyenne or L'Hôpital

There is a $\frac{1}{x-x_0}$ in front and one bound is fixed: that is the **definition of a derivative**. Families 5 have **nothing** in front and **both** bounds move.

4. Family 3 — Riemann sums (sums)**2 questions, 8–10 pts**

SIGNAL: $\lim_{n \rightarrow \infty} \sum_{k=1}^n (\dots)$ where k and n appear with the same power somewhere (k/n , k^2/n^2 , $k/n^2 \dots$)

Steps

1. Pull out $\frac{1}{n}$ in front. [2 pts]
2. Write everything left with $\frac{k}{n}$ only. If a lone n survives, it is NOT a Riemann sum.
3. Read $f(x)$: replace $\frac{k}{n}$ by x .
4. Say “ f continuous so integrable on $[0, 1]$ ”. [1 pt]
5. Limit $= \int_0^1 f(x) dx$.
6. Compute the integral (substitution, IPP, DES). [4–6 pts]

Example 3.1 (2023–24, 10 pts)

$$\sum_{k=1}^n \frac{k}{n^2} \text{Ln} \left(1 + \frac{k^2}{n^2} \right) = \frac{1}{n} \sum_{k=1}^n \frac{k}{n} \text{Ln} \left(1 + \left(\frac{k}{n} \right)^2 \right)$$

$f(x) = x \text{Ln}(1 + x^2)$, so the limit is $\int_0^1 x \text{Ln}(1 + x^2) dx$.

Sub $s = 1 + x^2$, $ds = 2x dx$:

$$= \frac{1}{2} \int_1^2 \text{Ln } s ds = \frac{1}{2} [s \text{Ln } s - s]_1^2 = \frac{1}{2} [(2 \text{Ln } 2 - 2) - (0 - 1)] = \boxed{\text{Ln } 2 - \frac{1}{2}}$$

Example 3.2 (Août 2020)

$$\sum_{k=1}^n \frac{k^2}{n^3} \text{Ln} \left(1 + \frac{k}{n} \right) = \frac{1}{n} \sum_{k=1}^n \left(\frac{k}{n} \right)^2 \text{Ln} \left(1 + \frac{k}{n} \right) \rightarrow \int_0^1 x^2 \text{Ln}(1 + x) dx$$

IPP with $u = \text{Ln}(1 + x)$, $v' = x^2$, $v = \frac{x^3}{3}$:

$$= \frac{\text{Ln } 2}{3} - \frac{1}{3} \int_0^1 \frac{x^3}{1+x} dx$$

DES by long division: $\frac{x^3}{1+x} = x^2 - x + 1 - \frac{1}{1+x}$, so $\int_0^1 \frac{x^3}{1+x} dx = \frac{1}{3} - \frac{1}{2} + 1 - \text{Ln } 2 = \frac{5}{6} - \text{Ln } 2$.

$$\text{Answer} = \frac{\text{Ln } 2}{3} - \frac{1}{3} \left(\frac{5}{6} - \text{Ln } 2 \right) = \boxed{\frac{2 \text{Ln } 2}{3} - \frac{5}{18}}$$

5. Family 4 — Riemann sums (products)**4 questions, 5 pts each**

SIGNAL: $\lim_{n \rightarrow \infty} \left(\prod_{k=1}^n a_{k,n} \right)^{1/n}$ or $\sqrt[n]{\prod \dots}$

Protocol (identical every time)

1. Pose $L =$ the limit.
2. $\text{Ln } L = \lim \frac{1}{n} \sum \text{Ln}(a_{k,n})$ (the $1/n$ exponent becomes the $\frac{1}{n}$ factor, the product becomes a sum). [2 pts]
3. Simplify inside: $\frac{n+k}{n} = 1 + \frac{k}{n}$, $\frac{n^2+k^2}{n^2} = 1 + (\frac{k}{n})^2$, $\text{Ln } e^a = a$.
4. Recognize $\int_0^1 f$ and compute (it is one of the 4 reference integrals). [2-3 pts]
5. **Come back:** $L = e^{\text{Ln } L}$ on a separate line. [2 pts]

Example 4.1 (Juil. 2021)

$$\sqrt[n]{\prod_{k=1}^n \frac{n+k}{n}} : \text{Ln } L = \lim \frac{1}{n} \sum \text{Ln}\left(1 + \frac{k}{n}\right) = \int_0^1 \text{Ln}(1+x) dx = 2 \text{Ln } 2 - 1$$

$$L = e^{2 \text{Ln } 2 - 1} = \boxed{4/e}.$$

Example 4.2 (Oct. 2022)

$$\sqrt[n]{\prod_{k=1}^n \frac{n^2+k^2}{n^2}} : \text{Ln } L = \int_0^1 \text{Ln}(1+x^2) dx = \text{Ln } 2 - 2 + \frac{\pi}{2}$$

$$L = e^{\text{Ln } 2 - 2 + \pi/2} = \boxed{2e^{\pi/2-2}}.$$

Examples 4.3 / 4.4 (Juin 2025, Juil. 2023)

$\text{Ln } e^a = a$ removes the exponential at once.

$$\left(\prod e^{\sqrt{(n-k)/n}}\right)^{1/n} : \text{Ln } L = \lim \frac{1}{n} \sum \sqrt{1 - \frac{k}{n}} = \int_0^1 \sqrt{1-x} dx = \frac{2}{3}, \quad L = \boxed{e^{2/3}}$$

$$\left(\prod e^{\sqrt{(n^2-k^2)/n^2}}\right)^{1/n} : \text{Ln } L = \int_0^1 \sqrt{1-x^2} dx = \frac{\pi}{4} \text{ (quarter circle)}, \quad L = \boxed{e^{\pi/4}}$$

6. Family 5 — limits of $\int_x^{2x}$

3 questions, 4-5 pts each

SIGNAL: nothing in front of the integral, **both** bounds move (x and $2x$, or x and $3x$). Then look where x goes.

F5a: $x \rightarrow 0^+$ or a finite number $\Rightarrow$ th. de la moyenne

1. Split the function as $\frac{h(t)}{t}$ (isolate $\frac{1}{t}$).
2. Say: h continuous, $\frac{1}{t}$ keeps a constant sign on $[x, 2x]$. [2 pts]
3. Exists $\xi_x \in [x, 2x]$: $\int_x^{2x} \frac{h(t)}{t} dt = h(\xi_x) \int_x^{2x} \frac{dt}{t} = h(\xi_x) \text{Ln } 2$.
4. $x \leq \xi_x \leq 2x \Rightarrow \xi_x \rightarrow 0$. [1 pt]
5. Answer $= h(0) \text{Ln } 2$. [2 pts]

Examples 5.1, 5.2

$$\int_x^{2x} \frac{e^{t+1}}{t} dt: h(t) = e^{t+1}, h(0) = e, \text{ limit} = \boxed{e \text{Ln } 2}.$$

$$\int_x^{2x} \frac{\sqrt{\sin t}}{t\sqrt{t}} dt: h(t) = \sqrt{\frac{\sin t}{t}} \rightarrow 1, \text{ limit} = \boxed{\text{Ln } 2}.$$

$$\text{Test p.16: } \int_x^{3x} \frac{\cos t}{t} dt, x \rightarrow 0^+: h = \cos t, h(0) = 1, \text{ limit} = \boxed{\text{Ln } 3}.$$

F5b: $x \rightarrow +\infty \Rightarrow$ critère de Cauchy

1. Say: $\int_1^{+\infty} (...) dt$ **converges** (give the reason). [2 pts]
2. Cauchy: $X_1 = x$ and $X_2 = 2x$ both go to $+\infty$, so $\int_{X_1}^{X_2} \rightarrow 0$. [2 pts]
3. Answer $= \boxed{0}$.

Example 5.3: $\lim_{x \rightarrow +\infty} \int_x^{2x} \frac{\cos t}{t} dt$

Converges by **test de Dirichlet**: $\cos t$ has a bounded primitive ($\sin t$), and $\frac{1}{t}$ decreases to 0. Then Cauchy $\Rightarrow \boxed{0}$.

Dirichlet signal: $\sin t$ or $\cos t$ times something decreasing to 0 ($\frac{1}{t}$, $\frac{1}{\sqrt{t}}$, $\frac{1}{t^2}$).

7. Family 6 — the guided exercise: Frullani

1 exercise, 15 pts

SIGNAL: $I = \int_0^{+\infty} \frac{f(ax) - f(bx)}{x} dx$, $f \rightarrow \ell$ at $+\infty$, $\int_0^1 \frac{f(t)}{t} dt$ convergent. Answer is always $\boxed{\ell \text{Ln } \frac{a}{b}}$.

Q1 — transform (4 pts): substitution + Chasles

Sub $t = ax$ in the first piece: $\frac{dx}{x} = \frac{dt}{t}$, bounds $a\varepsilon \rightarrow aX$. Same with $t = bx$. Then $\int_{a\varepsilon}^{aX} - \int_{b\varepsilon}^{bX}$: cut both at $b\varepsilon$ and aX (Chasles), the middle piece $\int_{b\varepsilon}^{aX}$ cancels, leaving $\int_{a\varepsilon}^{b\varepsilon} \frac{f(t)}{t} dt - \int_{aX}^{bX} \frac{f(t)}{t} dt$.

Q2 — $X \rightarrow \infty$ piece (5 pts): replace f by ℓ , bound the rest

$\int_{aX}^{bX} \frac{\ell}{t} dt = \ell \operatorname{Ln} \frac{b}{a}$ (this is the $\operatorname{Ln}(2x) - \operatorname{Ln} x$ trick). Difference: $\int_{aX}^{bX} \frac{f(t)-\ell}{t} dt$. Since $f \rightarrow \ell$, for $t \geq A$: $|f(t) - \ell| < \delta$, so $|\int| \leq \delta \operatorname{Ln} \frac{b}{a}$, as small as you want. Hence limit $= \ell \operatorname{Ln} \frac{b}{a}$.

Q3 — conclude (5 pts): $\varepsilon \rightarrow 0^+$ and $X \rightarrow \infty$

ε piece $\rightarrow 0$ by Cauchy (both bounds $\rightarrow 0$, integral convergent at 0). X piece $\rightarrow \ell \operatorname{Ln} \frac{b}{a}$ by Q2.

$$I = 0 - \ell \operatorname{Ln} \frac{b}{a} = \boxed{\ell \operatorname{Ln} \frac{a}{b}} \quad \text{With } a = 2, b = 3 : I = \ell \operatorname{Ln} \frac{2}{3}.$$

8. Family 7 — question de cours (proof by heart)

1 question, 7 pts, partial only

SIGNAL: “Montrer que f bornée est Riemann-intégrable ssi $\forall \varepsilon > 0, \exists P_\varepsilon : U(f; P_\varepsilon) - L(f; P_\varepsilon) < \varepsilon$ ”

 $\Rightarrow$ (integrable $\rightarrow$ small gap)

1. Integrable means $U(f) = L(f)$.
2. $U(f) = \inf_P U(f; P)$: pick P_1 with $U(f; P_1) < U(f) + \frac{\varepsilon}{2}$. $L(f) = \sup_P L(f; P)$: pick P_2 with $L(f; P_2) > L(f) - \frac{\varepsilon}{2}$. [1]
3. $P_\varepsilon = P_1 \cup P_2$. Refining lowers U , raises L . [1]
4. $U(f; P_\varepsilon) - L(f; P_\varepsilon) \leq U(f; P_1) - L(f; P_2) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$. [2]

 $\Leftarrow$ (small gap $\rightarrow$ integrable)

5. Always $L(f; P) \leq L(f) \leq U(f) \leq U(f; P)$ (sandwich), so $0 \leq U(f) - L(f) < \varepsilon$. [1]
6. True for every $\varepsilon > 0$, so $U(f) - L(f) = 0$: integrable. [2]

3 ideas carry the proof

$\frac{\varepsilon}{2} + \frac{\varepsilon}{2}$ split · refinement (add points: $U \downarrow, L \uparrow$) · sandwich $L(f; P) \leq L \leq U \leq U(f; P)$.

9. Toolbox for the “compute the integral” step**Intégration par parties (IPP)**

$\int u v' = u v - \int u' v$. **Signal:** Ln , arctan, arcsin alone, or polynomial $\times$ (Ln , e^x , $\sin$, $\cos$).

Choose u by **LIATE** priority: Ln , Inverse trig, Algebraic, Trig, Exp.

Trick: for $\int \operatorname{Ln}(1+x)$ take $v = 1+x$ (not x) so $v \cdot u' = 1$.

Changement de variable

Signal: you see $u(x)$ and $u'(x)$ together, e.g. $x \operatorname{Ln}(1+x^2)$ ($s = 1+x^2, ds = 2x dx$), or $\sqrt{1-x}$ ($s = 1-x$).

Always change the **bounds** too: $x = 0 \rightarrow s = 1, x = 1 \rightarrow s = 2$.

Symmetry trick (§6): $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$

Signal: $\int_0^{\pi/4} \operatorname{Ln}(1+\tan x) dx, \int_0^{\pi/2} \frac{\sin^m x}{\sin^m x + \cos^m x} dx$.

Apply $x \mapsto a+b-x$, the integral comes back, solve $I = c - I$.

Answers: $\frac{\pi}{8} \operatorname{Ln} 2$ and $\frac{\pi}{4}$.

Also: odd f on $[-a, a] \Rightarrow 0$; even $\Rightarrow 2 \int_0^a$.

Décomposition en éléments simples (DES)

Signal: a fraction $\frac{P(x)}{Q(x)}$ you cannot integrate directly.

deg $P \geq$ deg Q : long division first. Divide leading terms, multiply back, subtract, repeat. E.g. $\frac{x^3}{1+x} = x^2 - x + 1 - \frac{1}{1+x}$.

deg $P <$ deg Q : $\frac{1}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}$, multiply out, plug $x = a$ then $x = b$.

Add-and-subtract shortcut: $\frac{2x^2}{1+x^2} = 2 - \frac{2}{1+x^2}$.

Derivatives you keep forgetting

$(\sin x)' = \cos x, (\cos x)' = -\sin x, (\operatorname{Ln} x)' = \frac{1}{x}, (e^x)' = e^x, (\tan x)' = 1 + \tan^2 x$.

$(\sin^2 x)' = 2 \sin x \cos x = \sin 2x, (\sin(x^2))' = 2x \cos(x^2)$.

$(\arctan x)' = \frac{1}{1+x^2}, (\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$.

12-day plan (Bloc D part)

Day 1–2: F1 + F2 (HAUT–BAS, domain by segment). Day 3–4: F3 + F4 (Riemann, Ln -exp, the 4 integrals, IPP/DES). Day 5: F5 (moyenne / Cauchy). Day 6: Frullani once, then the 30-min test on p.16 of Bloc D. Last day: reread this sheet + F7 proof.